

HW 3 Converting to Standard Form practice

$$\begin{aligned} \text{maximize } & Z = 2x_1 + x_2 \\ \text{s.t. } & x_1 + x_2 \leq 1 \\ & x_1 \geq 0, x_2 \geq 0 \end{aligned}$$

0) Convert variables

1) Flip Sign (Always min)

2) Add Slack

$$\begin{aligned} \text{minimize } & Z' = -2x_1 - x_2 \\ & x_1 + x_2 + s_1 = 1 \\ & x_1, x_2, s_1 \geq 0 \end{aligned}$$

$$\text{if } x_1 \leq 0 \text{ then } x_1 = -x_1'$$

$$\text{if } x_1 \geq 2 \text{ then } x_1 = x_1' - 2$$

$$\text{if } x_1 \text{ free then } x_1 = x_1' - x_1''$$

Practice Complete attempting HW

1. Convert the following to Standard Form:

$$\begin{aligned} \text{maximize } & Z = 3x_1 + 5x_2 - 4x_3 \\ \text{s.t. } & 7x_1 - 2x_2 - 3x_3 \geq 4 \\ & -2x_1 + 4x_2 + 8x_3 = -3 \\ & 5x_1 - 3x_2 - 2x_3 \leq 9 \\ & x_1 \geq 1, x_2 \leq 7, x_3 \geq 0 \end{aligned}$$

$$\rightarrow x_1 = x_1' + 1 \quad x_2 = 7 - x_2' \quad x_3 = x_3'$$

$$(Z = -Z = -3x_1 - 5x_2 + 4x_3, \text{ sub \& drop constant})$$

$$\text{minimize } Z' = -3x_1' + 5x_2' + 4x_3'$$

$$\text{s.t. } 7x_1' + 2x_2' - 3x_3' - e_1 = 11$$

$$-2x_1' - 4x_2' + 8x_3' = -29 \text{ (multiply row by -1)}$$

$$5x_1' + 3x_2' - 2x_3' + s_3 = 25$$

$$x_1', x_2', x_3', e_1, s_3 \geq 0$$

$$\text{where } x_1 = x_1' + 1, x_2 = 7 - x_2', Z = 38 - Z'$$

2. Convert to Standard form

$$\checkmark \text{ Minimize } z = x_1 - 5x_2 - 7x_3 \quad \left[z' = x_1' - 5x_2' + 5x_2'' - 7x_3' - 7x_3'' \right]$$

$$\text{where: } z = z' + 2$$

$$\checkmark x_1 = x_1' - 2 \quad x_2 = x_2' - x_2'' \quad x_3 = x_3' - x_3''$$

$$\text{Subject to } 5x_1' - 10 - 2x_2' + 2x_2'' + 6x_3' - 6x_3'' = e_1 = 5$$

$$\left[5x_1' - 2x_2' + 2x_2'' + 6x_3' - 6x_3'' - e_1 = 15 \right]$$

$$3x_1' - 6 + 4x_2' - 4x_2'' - 9x_3' + 9x_3'' = 3$$

$$\left[3x_1' + 4x_2' - 4x_2'' - 9x_3' + 9x_3'' = 9 \right]$$

$$7x_1' - 14 + 3x_2' - 3x_2'' + 5x_3' - 5x_3'' + s_3 = 9$$

$$\left[7x_1' + 3x_2' - 3x_2'' + 5x_3' - 5x_3'' + s_3 = 23 \right]$$

$$x_1', x_2', x_3', e_1, s_3 \geq 0$$

3. Convert to standard form

$$\times \text{ Maximize } z = 6x_1 - 3x_2$$

$$\text{Minimize } z' = -6x_1 + 3x_2 \rightarrow z' = -90 + 6x_1' + 45 - 3x_2'$$

$$\left[z = z' + 45 \right]$$

$$\left[z' = 6x_1' - 3x_2' \right]$$

$$\text{where } \cancel{x_1 = 15 - x_1'} \quad \left[x_1 = 15 - x_1' \right] \quad \left[x_2 = 15 - x_2' \right]$$

$$\text{Subject to } 30 - 2x_1' + 75 - 5x_2' - e_1 = 10$$

$$\left[-2x_1' - 5x_2' - e_1 = -95 \right]$$

$$45 - 3x_1' + 30 - 2x_2' + s_2 = 40$$

$$\left[-3x_1' - 2x_2' + s_2 = -35 \right]$$

$$\left[x_1', x_2', e_1, s_2 \geq 0 \right]$$

4. Determine Basic Solutions from Standard Form

$$x_1 + x_2 \leq 5$$

$$x_1 + 2x_2 \leq 6$$

$$x_1 \geq 0 \quad x_2 \geq 0$$

$$x_1 + x_2 + s_1 = 5$$

$m=2$ equations

$$x_1 + 2x_2 + s_2 = 6$$

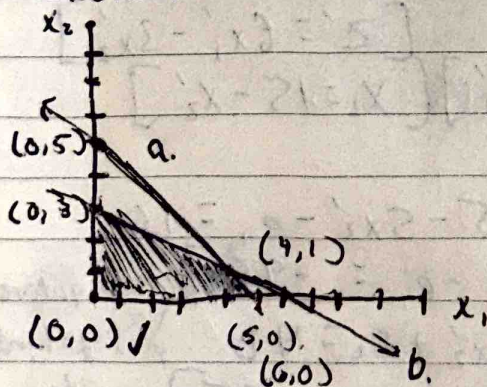
$n=4$ variables

$$x_1, x_2, s_1, s_2 \geq 0$$

Every pair set to 0 (x_1, x_2)

- | | | |
|---|---|--------------------------------|
| 1 | $x_1, x_2 \therefore s_1 = 5, s_2 = 6$ | $(0, 0) \checkmark$ |
| 2 | $x_1, s_1 \therefore x_2 = 5, s_2 = -4$ | $(0, 5) \times \{s_2 \leq 0\}$ |
| 3 | $x_1, s_2 \therefore x_2 = 3, s_1 = 2$ | $(0, 3) \checkmark$ |
| 4 | $x_2, s_1 \therefore x_1 = 5, s_2 = 1$ | $(5, 0) \checkmark$ |
| 5 | $x_2, s_2 \therefore x_1 = 6, s_1 = -1$ | $(6, 0) \times \{s_1 \leq 0\}$ |
| 6 | $s_1, s_2 \therefore x_1 = x_2 = 1$ | $(4, 1) \checkmark$ |

Sketch



$$\text{line } x_1 + x_2 = 5 = a$$

$$x_1 + 2x_2 = 6 = b$$

The 4 feasible basic solutions are the 4 corners of the solution space.